Name
Date
Course

Factoring Algebraic Expressions (Part I)

Factor out the greatest common factor.

① $28x^5 - 8x^3 + 12x^2 = \boxed{4x^2(7x^3 - 2x + 3)}$

② $21y^4 - 27y^3 + 15y = \boxed{3y(7y^3 - 9y^2 + 5)}$

③ $-50z^6 + 40z^4 - 15z^3 = \boxed{-5z^3(10z^3 - 8z + 3)}$

④ $4x^4 - 8x^3 + 20x^2 = \boxed{4x^2(x^2 - 2x + 5)}$

⑤ $-12y^4 + 33y^3 - 15y^5 = \boxed{-3y^3(4y - 11 + 5y^2)}$

⑥ $25z^6 - 40z^5 - 15z^3 = \boxed{5z^3(5z^3 - 8z^2 - 3)}$

⑦ $6h^6 - 14h^2g = \boxed{2h^2(3h^4 - 7g)}$

⑧ $24a^5b^3 - 18a^2 - 36a^3b^2 = \boxed{6a^2(4a^3b^3 - 3 - 6ab^2)}$

⑨ $-6c^7e^3 - 9c^2e + 15e^3 = \boxed{-3e(2c^7e^2 + 3c^2 - 5e^2)}$

⑩ $12h^6g^3 - 20h^2g^2 = \boxed{4h^2g^2(3h^4g - 5)}$

⑪ $14a^5b^3 - 21a^2b - 42a^3b^2 = \boxed{7a^2b(2a^3b^2 - 3 - 6ab)}$

⑫ $-24c^4e^3 - 9c^2e + 15c^3 = \boxed{-3c^2(8c^2e^3 + 3e - 5c)}$

⑬ $72x^4y^3z^4 + 27x^2y^3z - 81x^3y^2z^6 + 45x^5y^6z^3$

$\boxed{9x^2y^2z(8x^2yz^3 + 3y - 9xz^5 + 5x^3y^4z^2)}$

⑭ $63x^5y^3z^4 + 35x^2yz^5 - 56x^3y^2z^6 + 28x^{10}y^6z^3$

$\boxed{7x^2yz^3(9x^3y^2z + 5z^2 - 8xyz^3 + 4x^8y^5)}$

Factor the quadratic expressions.

⑮ $x^2 + 8x + 15 = \boxed{(x+3)(x+5)}$

⑯ $y^2 + 6y + 5 = \boxed{(y+1)(y+5)}$

⑰ $z^2 + 7y + 12 = \boxed{(z+3)(z+4)}$

⑱ $x^2 + 8x + 12 = \boxed{(x+2)(x+6)}$

⑲ $y^2 + 4y + 4 = \boxed{(y+2)(y+2)}$

⑳ $x^2 + 9x + 18 = \boxed{(x+6)(x+3)}$

㉑ $y^2 + 4y + 3 = \boxed{(y+1)(y+3)}$

㉒ $z^2 + 2z + 1 = \boxed{(z+1)(z+1)}$

㉓ $x^2 + 12x + 27 = \boxed{(x+3)(x+9)}$

㉔ $y^2 + 17y + 72 = \boxed{(y+9)(y+8)}$

㉕ $z^2 + 15z + 56 = \boxed{(z+7)(z+8)}$

㉖ $z^2 + 3z - 10 = \boxed{(z-2)(z+5)}$

㉗ $y^2 - 3y - 28 = \boxed{(y+4)(y-7)}$

㉘ $x^2 - 15x + 54 = \boxed{(x-6)(x-9)}$

㉙ $z^2 - 7z - 44 = \boxed{(z-11)(z+4)}$

㉚ $y^2 + 5y + 6 = \boxed{(y+2)(y+3)}$

(31) $x^2+6x+8 = \boxed{(x+2)(x+4)}$

(32) $z^2+4z-21 = \boxed{(z-3)(z+7)}$

(33) $y^2+7y+12 = \boxed{(y+3)(y+4)}$

(34) $x^2-5x-50 = \boxed{(x-10)(x+5)}$

(35) $z^2-8z+15 = \boxed{(z-3)(z-5)}$

(36) $x^2+7x-30 = \boxed{(x+10)(x-3)}$

(37) $z^2-12z+36 = \boxed{(z-6)(z-6)}$

(38) $y^2-8y-9 = \boxed{(y-9)(y+1)}$

(39) $x^2+11x+30 = \boxed{(x+5)(x+6)}$

(40) $z^2+19z-20 = \boxed{(z+20)(z-1)}$

(41) $y^2+10y-24 = \boxed{(y+12)(y-2)}$

(42) $x^2-x-12 = \boxed{(x-4)(x+3)}$

(43) $z^2-10z+25 = \boxed{(z-5)(z-5)}$

(44) $y^2+6y-7 = \boxed{(y+7)(y-1)}$

(45) $x^2+24x+80 = \boxed{(x+20)(x+4)}$

(46) $z^2-13z+40 = \boxed{(z-5)(z-8)}$

(47) $2y^2+7y+3 = \boxed{(y+3)(2y+1)}$

(48) $3x^2+x-4 = \boxed{(3x+4)(x-1)}$

(49) $5z^2-6z+1 = \boxed{(5z-1)(z-1)}$

(50) $2y^2+7y+6 = \boxed{(2y+3)(y+2)}$

(51) $3x^2-10x-8 = \boxed{(3x+2)(x-4)}$

(52) $2y^2-7y-15 = \boxed{(2y+3)(y-5)}$

(53) $3x^2-4x-4 = \boxed{(3x+2)(x-2)}$

(54) $6z^2-13z-28 = \boxed{(2z-7)(3z+4)}$

(55) $4y^2-17y+15 = \boxed{(4y-5)(y-3)}$

(56) $3x^2+10x+3 = \boxed{(3x+1)(x+3)}$

(57) $5z^2+18z-8 = \boxed{(5z-2)(z+4)}$

(58) $7y^2-12y+5 = \boxed{(7y-5)(y-1)}$

(59) $11x^2+45x+4 = \boxed{(11x+1)(x+4)}$